# Architecture Document

## 1. APPLICATION ARCHITECTURE

### Selected Architecture: Event-Driven Architecture (Data Pipeline)

### Detailed Explanation:

The GLOF monitoring system is designed using an **Event-Driven Architecture**, where each stage of the pipeline triggers the next stage automatically without manual intervention. The system processes large-scale satellite and DEM datasets sequentially, forming a fully automated computational pipeline.

According to the implemented methodology, the system follows a structured execution flow where each module generates outputs that act as inputs for the subsequent stage. This ensures continuous processing and scalability.

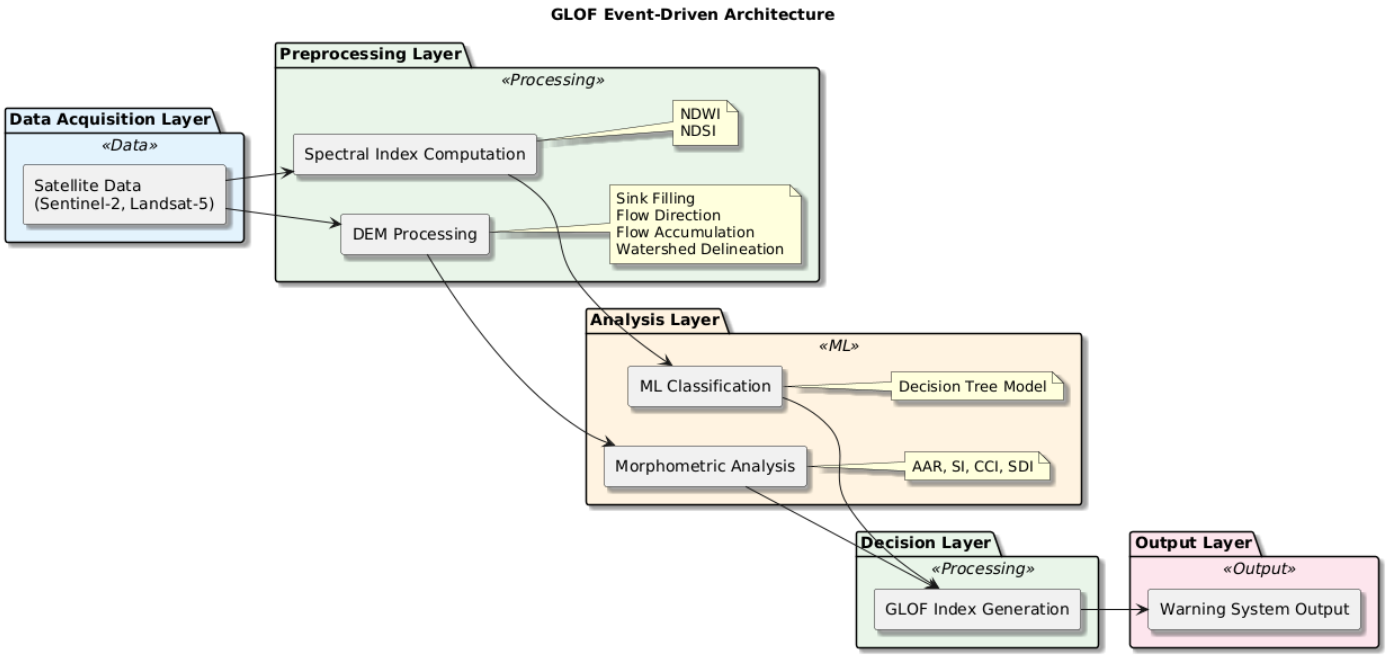
The workflow consists of:

- Data acquisition using **Sentinel-2 (5-day revisit)** and **Landsat-5 historical datasets (1988–2023)**
- DEM-based hydrological preprocessing (sink filling, flow direction, accumulation)
- Spectral index computation (NDWI, NDSI)
- Machine Learning classification using a **Decision Tree model**
- Morphometric parameter computation (AAR, SI, CCI, SDI)
- GLOF Index generation
- Final warning system output

This architecture ensures:

- Automation of the entire pipeline
- Elimination of manual intervention
- High scalability for large datasets
- Real-time or near-real-time hazard assessment

## Architecture Diagram (Event Flow)



## 2. DATABASE DESIGN

### 2.1 ER Diagram

**Explanation:**

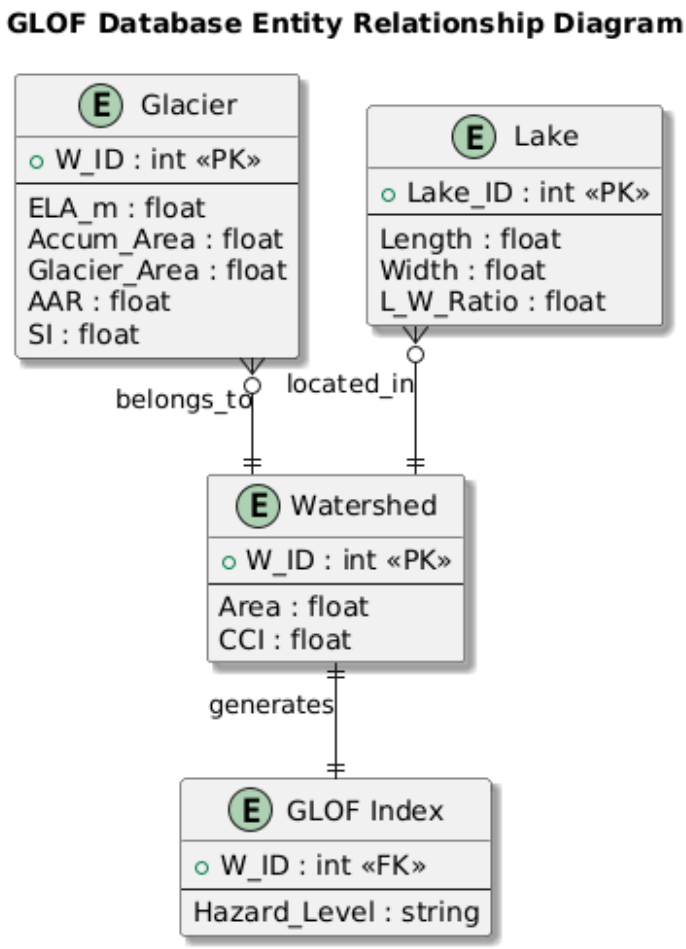
The system utilizes a **Tri-Matrix Data Architecture**, where spatial and computational outputs are stored in structured datasets.

The three primary entities are:

1. **Glacier Parameters Matrix**
   - Contains glacier-related metrics such as:
     - Accumulation Area Ratio (AAR)
     - Snout Index (SI)
     - Glacier Area
2. **Glacial Lake Parameters Matrix**
   - Contains lake-related morphometric data:
     - Length-to-Width Ratio
     - Catchment Compactness Index (CCI)
3. **GLOF Index Matrix**
   - Final output dataset containing:
     - Hazard classification (Very Low → Very High)

Each **Micro-Watershed (W_ID)** acts as the linking key across all datasets.

**ER Diagram**



## 2.2 Schema Design

## Explanation:

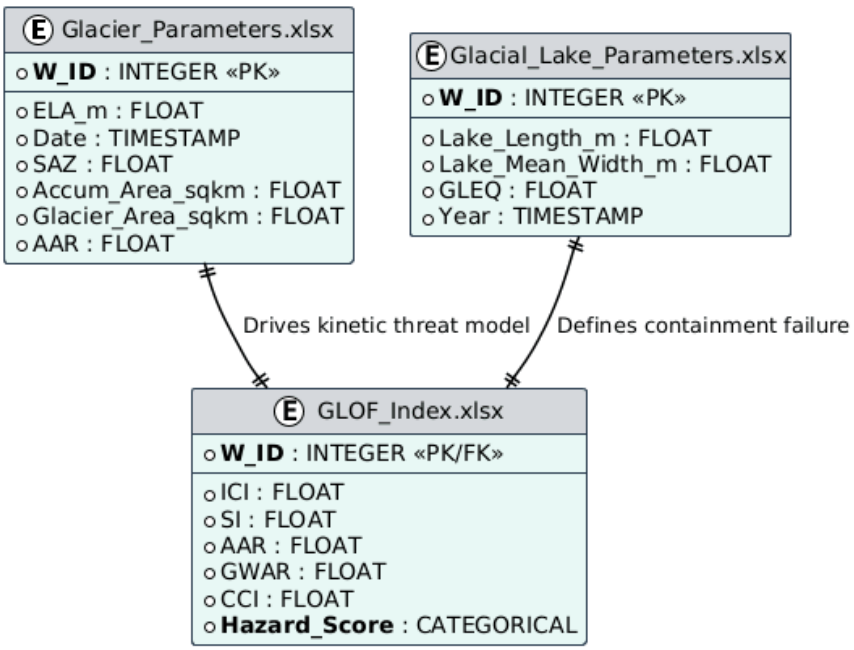
The system stores processed data into structured spreadsheet-based datasets (.xlsx format), enabling easy integration with analytical models and decision-making systems.

The schema consists of:

- **Glacier Parameters Table**
  - W_ID (Primary Key)
  - AAR
  - SI
- **Lake Parameters Table**
  - Lake_ID
  - L/W Ratio
  - CCI
- **GLOF Index Table**
  - W_ID (Foreign Key)
  - Hazard Level

These tables are interconnected through the watershed identifier (W_ID), enabling unified analysis.

## Database Schema Diagram




### 3. DATA EXCHANGE CONTRACT

### 3.1 Frequency of Data Exchange

- **Sentinel-2 Data:** Every 5 days (high temporal resolution)
- **Landsat-5 Data:** Historical dataset (1988–2023) used for long-term analysis

## 3.2 Data Sets

The system processes multiple datasets:

- Satellite imagery (.tif format)
- DEM data (ASTER GDEM)
- Spectral indices (NDWI, NDSI)
- Hydrological outputs (Flow accumulation, watersheds)
- Morphometric parameters (AAR, SI, CCI, SDI)
- Final GLOF Index dataset

### 3.3 Mode of Exchange

- Internal pipeline data flow between modules
- File-based exchange (.xlsx, GeoJSON, raster data)
- Future scope includes API-based integration

## Data Flow Diagram (DFD)

